Steve Andreas Immanuel

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Experiences

AI Engineer — **TelePIX** — *Feb 2024 - Current*
- Researched and developed a novel algorithm for multispectral satellite image compression
- Proposed novel methodology for few-shot segmentation in remote sensing domain
- Implemented LLM framework with RAG and multi-step reasoning capabilities for in-house chatbot

AI Researcher — **VLI (Sejong University)** — *Feb 2022 - Feb 2024*
- Proposed novel architecture for video question answering task with low computational complexity
- Implemented reinforcement learning policy gradient algorithm for novel video captioning model
- Developed many machine learning pipelines for various tasks, e.g., video captioning, keypoint detection, object detection, etc.

Software Engineer (Contract) — **Mekari** — *Nov 2021 - Feb 2022*
- Cut down API latency by 50% by refactoring and improving old legacy code
- Designed a new public API ecosystem to be consumed by many other third parties
- Developed a new security mechanism to perform secure computation for sensitive data in an untrusted environment

Backend Engineer — **SejutaCita** — *Feb 2021 - Aug 2021*
- Developed a payment system for an application that has been used by more than 600k users
- Implemented a publish-subscribe system for internal communication in a microservices environment
- Refactored and created an end-to-end testing mechanism that is easily maintainable and extensible

Selected Publications

Tackling Few-Shot Segmentation in Remote Sensing via Inpainting Diffusion Model
Steve Andreas Immanuel, Woojin Cho, Junhyuk Heo, Darongsae Kwon
ICLR 2025 Workshop on Machine Learning for Remote Sensing (Oral)

Neural Compression for Multispectral Satellite Images
Woojin Cho*, **Steve Andreas Immanuel***, Junhyuk Heo, Darongsae Kwon
NeurIPS 2024 Workshop on Machine Learning and Compression

Learnable Prompt for Few-Shot Semantic Segmentation
Steve Andreas Immanuel, Hagai Raja Sinulingga
CVPR 2024 Workshop on Learning with Limited Data

Lightweight Recurrent Cross-modal Encoder for Video Question Answering
Steve Andreas Immanuel, Cheol Jeong
Knowledge-Based Systems 2023

Improving Distinctiveness in Video Captioning with Text-Video Similarity
Vania Velda, **Steve Andreas Immanuel**, Willy Fitra Hendria, Cheol Jeong
Image and Vision Computing 2023

Education

Master of Intelligent Mechatronics — **Sejong University** — *Mar 2022 - Feb 2024*
- GPA: 4.50 / 4.50
- Main research topics: vision-language understanding, NeRF, computer vision
- Curated and taught materials for artificial intelligence and machine learning courses for undergraduate students

Bachelor of Computer Science — **Institut Teknologi Bandung** — *Aug 2017 - Jul 2021*
- GPA: 3.96 / 4.00
- Earned “Mahasiswa Prestasi” outstanding award for having a perfect GPA during the freshman year
- Taught the programming course for first-year students

Awards

- **1st Winner** of AI4EO Map Your City Challenge (2024)
  - Building multimodal classifier to identify building age from top-view satellite imagery
- **4th Winner** of OpenEarthMap Land Cover Mapping Few-Shot Challenge (2024)
  - Developing segmentation model and algorithm for few-shot settings

Selected Projects

- **QuickTorch**
  - PyTorch framework for quick development for general machine learning, complete with distributed training support
  - Tech Stack: **PyTorch, Optuna**
- **Torch Cosine Annealing**
  - Python library that implements cosine annealing scheduler introduced in SGDR paper with additional features
  - Tech Stack: **PyTorch**
- **TrackMe**
  - Integrated LINE chatbot and mobile app to track user location autonomously and provide notifications via chat
  - Tech Stack: **Flutter, MongoDB, InfluxDB, Redis**

Peer Reviews

- IEEE Transactions on Geoscience and Remote Sensing
- Knowledge-Based System
- Image and Vision Computing

Skills

Programming: Python, Javascript, Java, C++, C#, C, PHP, SQL, NoSQL

Technologies: Torch, Tensorflow, NodeJS, Unity, Docker, Flutter, Firebase, AWS, GCP, Azure, Qt, CI-CD, and many more

Languages: Indonesian (Native), English (Expert), Korean (Beginner)